

The Agent Graph Runtime

A Unified Model for Static, Dynamic, and Hybrid Execution

Rashid Azarang

v1.0 · 22 August 2026

Internal preprint. Six sealed certification epochs (r1–r6) were attempted; none produced a certifiable review, and the two that reached adjudication (r3, r6) returned noncompliant on producer conduct. Every record is preserved and disclosed in Appendix A. By dated author decision (2026-08-14), certification is no longer a release precondition; disclosure of the six-epoch record, incorporation of the r6 uncertified blind-review dispositions, and the author’s explicit release act are. That act occurred on 2026-08-22 (Decision A1, recorded with the six-epoch provenance in the public repository); this version is that release.

Abstract

Agent systems are often classified as static workflows, dynamic planners, or hybrid arrangements. Those labels conflate when a graph is constructed, where it becomes immutable and addressable, how it is authored, and what executes it. This paper defines an agent graph runtime model in which those concerns are separately specified lifecycle coordinates over a constrained feasible region and a shared executable representation and execution substrate. Static, dynamic one-shot, staged-direct, and staged-scaffold configurations are reference points in that region, not necessarily distinct runtimes. The model does not assert that every Cartesian combination is meaningful or implementable.

The model separates construction, lowering, qualification, freezing, requalification, admission, execution, and evidence. It also separates properties that are easily confused: executable graph identity, execution environment identity, output reproducibility, plan adequacy, and outcome correctness. MENTU is the paper’s author-checked architectural instantiation of the strict-DAG admitted agent-graph profile. At the current architecture snapshot, `MentuExecutionGraphCore` owns the normalized executable, lowering, qualification, admission, lifecycle evidence, and DAG scheduler; engine adapters supply host effects. Mentu also retains persistent recipe and interpreted workflow compatibility profiles with different assurance contracts. One current cross-profile boundary remains explicit: a generated persistent projection can be rediscovered by legacy recipe dispatch without a named adoption transition. The unmodified projection has no `auto_mode` marker; for either CLI-auto Boolean, ordinary dispatch constructs the raw runner and enters legacy authority. More generally, every cell of the two-by-two recipe-auto/CLI-auto matrix that passes its applicable precondition does so. The sole conditional committed-recipe existence gate is not Core qualification or admission. The product integration therefore does not yet enforce the model’s projection-is-not-adoption rule. The admitted Core path remains that instantiation; the exception is not generalized to that path or hidden as uniform product-wide conformance. A separate exact-archive, author-side run at the current source pin passes 25/25

targeted tests across 13 filters, including lowering parity, canonicalization, admission identity and drift, saved-plan execution, and an admitted vertical slice. The suite does not execute an ordinary static recipe and a generated graph through one common Core admission route, so it is not evidence of product-wide executed cross-origin parity.

A frozen historical C30 pilot is reported as a mechanical invariant exercise, not a treatment comparison. It reports identity equality for all five qualified persistent projections. Two of those projections were not selected for execution after contemporaneous human semantic rejection; the remaining three attempted saved-plan transitions all reached admission and retained identity. There were zero exclusions *within each reported attempted transition*, not zero attrition across the full projection-to-execution funnel. It also reports zero *recorded* graph-planner dispatches during those three executions and sealed-manifest integrity; three historical C30-derived controls and one separate current-Core control exercise their named detector paths. The same pilot retains negative evidence: mechanically qualified plans failed human semantic review, and a mechanically successful artifact later required semantic correction. No matched static or dynamic observations and no unchanged-state repetition series exist, so comparative superiority, amortization, and generality claims are excluded. The v0.9.5 return reported 95 literature records and 44 review findings, including 14 majors; independent audit found five copy-contaminated RFC author fields and only 34 honest matches for that version’s 37 citations. The v0.9.6 full raw gate failed. The v0.9.7 submitted files pass their structural validator and its blind review reports 21 findings, including six majors, but trace audit invalidates the run’s process claims and independent source audit finds an RFC 2748 first-author mismatch. The v0.9.8 attempt correctly stopped before either review when it could not produce an independently observed trace audit. These returns are bounded advisory or instrument-diagnostic evidence, not literature certificates or release authorization. Novelty remains uncertified. Fresh review of this exact PDF and supplement, plus a detached supplied-export audit, remains a release gate.

Keywords: agent runtime; execution graph; artifact admission; content-addressed execution; provenance; planner–executor separation

1 Introduction

An agent that performs consequential work has at least two conceptually different problems. It must decide what work is to be done, and it must enact that decision under bounded authority. In small demonstrations both problems can be hidden inside one prompt or one tool loop. In a systems setting they create different artifacts, failure modes, and review obligations. A planner may construct an unsuitable graph even when the executor correctly enforces every mechanical contract. Conversely, a suitable plan may be executed under the wrong authority, against changed state, or through a path that cannot bind its effects back to the inspected artifact.

The familiar labels *static*, *dynamic*, and *hybrid* are too coarse to locate these distinctions. “Dynamic” may mean task-time graph construction, runtime choice among already-declared branches, or mutation of the topology during execution. “Hybrid” may mean only that construction and execution occur in different invocations, or it may imply a second runtime. “Workflow” adds another ambiguity: it can denote a business process, an authoring notation, a script interpreted at runtime, or the lowered executable structure. Treating those meanings as synonyms turns an architectural boundary into a naming preference.

This paper takes the *agent graph lifecycle* as the unit of analysis. It asks which transformations and authority transitions connect an authored source to an effectful run and its evidence. The central thesis is deliberately narrow:

One execution substrate can support multiple graph-lifecycle policies. A system is unified at this level when different construction and freeze policies lower to a shared executable representation and run through common execution machinery. This does not require their authoring interfaces or authority-entry paths to be identical.

The paper contributes four bounded results. First, it supplies a construct model and a policy vector that place reference static, dynamic, and staged configurations in one space. Second, it states invariants for identity, qualification, freeze, admission, execution, and evidence, including a scope theorem that marks the edge of one frozen graph epoch. Third, it maps those constructs to current committed MENTU source and identifies an extracted admitted execution-graph Core with a public API, differentiated compatibility profiles, and one unresolved projection-adoption transition. Fourth, it reports a historical C30 invariant-exercise pilot, plus disposable seeded detector controls, while preserving the pilot’s null and negative findings.

These are architecture, implementation, formal-implication, and pilot observation claims. They are not evidence that staged execution improves task success, cost, latency, maintenance, or regression rates. They do not imply deterministic model output, plan adequacy, or outcome correctness. They do not establish generality beyond MENTU, and the historical C30 epoch is not pooled with the current architecture epoch. The internal version makes no priority or exclusive novelty claim. It likewise makes no claim that MENTU is globally first, best, or superior to every other agent runtime.

2 Related work and discriminant positioning

Review status. The exact-input v0.9.5 Operon return contains 95 records—59 retained, 30 background, three counterexamples, and three exclusions—and a separate blind context returned 44 findings, including 14 major findings. Its closed-tree, input-binding, isolation, and file-open validators pass. That process result does not establish bibliographic correctness. Independent audit found five RFC entries whose author field was copy-contaminated with “The Bazel Authors,” only 34 honest entity-level matches for the *v0.9.5 manuscript’s* 37 bibliography entries, and a live Kubernetes page substituted for the pinned bytes cited here. Two LangGraph documents and the cited 2019 in-toto paper have no one-to-one registry row. A release crosswalk is therefore impossible without fabrication, which this paper refuses.

The compliant raw return is retained as bounded discovery and blind-review evidence. Its mechanism coding is advisory because the search report itself records that close full-text reading was not completed. Earlier independently verified citation records remain claim-specific inputs, but neither an earlier review nor a same-work preprint silently certifies the source version cited in this revision. No corpus count, matrix blank, or “closest found” judgment establishes priority, exhaustive coverage, or a unique conjunction. The v0.9.5 blind review recommends major revision. The later v0.9.7 repository projection has a 43/43 cite-key-to-bibliography-key identity crosswalk, but that does not certify rendered author, venue, pagination, or publisher fields. Its Operon session is non-certifying under trace audit despite a green structural validator. `novelty_certified` remains false the flag is read from the repository artifact `results/citation-verification-v098.json` (sha256 62a80d37...), named here because that artifact sits outside any blind-review surface (finding BR9-007) and fresh literature verification and blind review of this exact PDF remain required for release.

Design-space and workflow precedents. The method of separating apparently monolithic systems into design coordinates is established. Build Systems à la Carte factors build systems into components and uses that factorization to compare designs [?]. We adopt that design-space discipline without assuming a full Cartesian product or claiming the move as novel. Pegasus distinguishes an abstract, resource-independent workflow from the executable workflows produced while mapping it to distributed infrastructure [?]. This is a clear precedent for separating source intent from executable form. Petri-net workflow soundness is defined over a net together with its firing semantics, not inferred from a topology label alone [?]. We therefore do not equate “workflow” with “DAG.”

Graph runtime, persistence, and compatibility. LangGraph’s official documentation specifies a graph runtime, checkpoint-based persistence, and an explicit backward-compatibility contract [?, ?, ?]. These are positive precedents for stateful graph execution, durable checkpoints, and version-evolution policy. This paper treats those capabilities as established runtime mechanisms and separately specifies its discriminants for executable identity, lifecycle freeze, and pre-effect artifact admission.

Define-by-run, tracing, and staged graph extraction. Dynamic construction followed by extraction of an executable graph is an established systems pattern, not a contribution of this paper. TensorFlow Eager combines imperative execution with a just-in-time tracer that translates Python functions into executable dataflow graphs [?]. That precedent directly occupies construction/freeze coordinates adjacent to the staged policies here. The remaining distinction is narrower: whether the extracted executable is independently addressable across invocations and whether a pre-effect transition binds its identity, qualification, authority, and concrete run context. The paper therefore claims a lifecycle-and-admission synthesis, not invention of define-by-run, graph tracing, staging, caching, or graph reuse.

Planning, acting, and agent control. Planning and acting have long been treated as separate but interacting problems [?]. Hierarchical task-network and partial-order planning supply established formalisms for decomposition and least-commitment action structure [?, ?]. In this paper, that action structure is the candidate graph g ; a saved *plan* is the addressable lifecycle record z that may bind g , its lowering, qualification context, and executable identity.

Agent systems occupy several points in the lifecycle region. ReAct interleaves reasoning and action [?]. LLMCompiler separates a function-calling planner, task-fetching unit, and executor, but its task graph is an in-process structure rather than the cross-invocation, content-addressed freeze used here [?]. DSPy separates declarative language-model modules from compilation into an optimized program; that is a strong authoring/lowering precedent, not evidence of a run-specific admission receipt [?]. AFlow searches over code-represented workflows using execution feedback [?], while ADaPT decomposes on executor failure [?]. These works establish interleaved control, planner–executor separation, compilation, and adaptive construction. GPTSwarm represents language-agent systems as optimizable graphs [?]; Open Agent Specification defines one portable agent description across several runtimes [?]; and Compiled AI separates generation and validation from later execution [?]. Xu et al.’s execution-time EvoMAS marks the other boundary: graph construction can continue during execution [?]. Our discriminants ask narrower systems questions: whether an executable is addressable across invocations, whether inspected identity survives an authority transition, and whether topology change starts a new admitted epoch. The Common Workflow Language supplies

a portable declarative workflow standard with explicit runtime-environment descriptions and execution across multiple implementations [?]. It is therefore a direct precedent for separating authoring syntax from portable execution, while this paper separately studies identity-bearing admission of an agent graph.

Stored plans, identity, and replay. PLANEX already combined stored generalized plans, pre-condition monitoring, and re-entry to planning on failure [?]; staged construct-check-execute is therefore not new in isolation. Nix demonstrates input-derived identity over declared build inputs, while reproducible-build work distinguishes that identity from bit-identical outputs [?, ?]. Those precedents narrow the identity contribution: this paper’s move is to state the placement rule for an agent executable and residual run environment, then keep plan adequacy and outcome correctness as separate non-implications. Durable Functions gives deterministic event-history replay semantics for stateful orchestration [?]; Temporal records a durable event history and replays workflow code to restore execution state after failure [?]. The present frozen-graph claim is different: it constrains graph construction during one admitted epoch and does not promise replay-equivalent effects.

Bazel’s Remote Execution API content-addresses declared actions and uses action digests for cache lookup [?]. OxyMake is a close identity-and-evidence neighbor in the bounded review: it specifies a content-addressed plan of record, append-only audit state, and programmatic gate approval [?]. The present model asks whether whole-executable identity, mechanical qualification, authority envelope, and concrete run context are jointly bound by a pre-effect receipt. This paper has not evaluated OxyMake conjunct-by-conjunct against that question and makes no claim that it lacks any conjunct. The comparison names an inspected neighbor; it does not assert structural uniqueness or exhaustive priority.

Admission and evidence precedents. “Admission control” conventionally includes resource and QoS decisions made through policy decision and enforcement points [?]. We use *artifact admission* for a narrower pre-effect authority transition that binds an executable identity to a run context; it is not a network resource reservation claim. Usage-control models provide a further analogue for rechecking mutable attributes and continuing authorization [?]. Kubernetes admission explicitly includes mutating and validating hooks [?]. Artifact admission here is the narrower non-mutating case: a mutation creates a new executable that must return to qualification before it can receive authority. Proof-carrying code places a producer-supplied proof against a receiver-defined safety policy before execution [?]. It is a strong pre-execution policy analogue, not an equivalence: this paper’s mechanical qualification does not claim the semantic force of a safety proof. Runtime monitoring results also bound what a particular enforcement mechanism can guarantee [?].

Supply-chain verification is an even closer operational neighbor. SLSA specifies verification of artifact provenance against trusted builders and expected parameters [?]; Sigstore supplies signing and transparency infrastructure [?], while its policy controller validates signatures and attestations during Kubernetes admission [?]. Gatekeeper provides policy-as-code validation and mutation operations around Kubernetes admission [?]. These systems establish pre-use artifact verification, signature/attestation policy, and admission enforcement. The paper does not claim they lack its conjuncts; it asks a different synthesis question about binding a complete agent executable, its qualification, and its concrete run context to authority on a shared execution substrate.

in-toto uses signed layouts, authorized functionaries, and link metadata to verify a software supply-chain history before artifact acceptance [?]. It is a close authority-and-evidence analogue, but does

not itself supply the shared agent-graph execution substrate or authoring-coordinate model defined here. PROV-DM provides a standard vocabulary for entities, activities, agents, derivation, and responsibility [?]. That vocabulary supports lineage, but delegation and responsibility records do not enact artifact admission, and provenance alone does not establish executable identity, output reproducibility, plan adequacy, or outcome correctness.

This revision does not pretend to be an exhaustive orchestrator survey. Dagster, Prefect, Argo, and Airflow remain named coverage candidates for the fresh literature pass; no absence or uniqueness inference depends on them.

The paper uses supply-chain vocabulary narrowly. An *admission receipt* is an authorizing record emitted before host effects and bound to the admitted artifact and context. A sealed manifest or activation record is a retrospective *attestation*: evidence about what was activated or produced. Neither term implies a digital signature unless the realization supplies one, and retrospective evidence cannot substitute for pre-effect authority.

Recent agent-governance systems enforce or monitor policies around action: AgentSpec supplies customizable runtime enforcement [?]; TrustAgent places safety checks before, during, and after planning [?]; MI9 models continuous authorization [?]; and SARC proposes pre-action architectural governance [?]. These are enforcement neighbors, not evidence that qualification and artifact admission are synonymous. The agent-provenance literature studies retrospective execution evidence [?], whereas the executable graph here is a prospective control object whose later evidence is separately sealed.

Operational discriminants. We position a mechanism only through five questions: (1) when graph structure is constructed; (2) whether an immutable executable becomes independently addressable across invocations; (3) whether authoring method is specified separately from runtime authority; (4) whether distinct origins converge on the same runner and scheduling semantics; and (5) whether a pre-effect transition binds executable, qualification, authority, and run context. These predicates are not a novelty score. The contribution claimed here is the explicit lifecycle synthesis, its source-grounded mapping to MENTU, and a mechanical invariant exercise—not graphs, planning before action, lowering, policy checks, content identity, or provenance in isolation.

Table 1 replaces an earlier asymmetric feature matrix. It records only positive mechanism precedents supported by the cited primary or official source; it does not code uninspected absence. The final row is the paper’s current implementation mapping, not a literature claim.

3 Agent graph runtime model

Constructs

An *agent graph* is a typed control structure

$$G = (V, E, \lambda, \kappa, \sigma_g),$$

where V is a set of action or decision nodes, E is a set of typed coordination relations, λ binds node behavior, κ carries contracts and authority requirements, and σ_g defines graph-declared scheduling semantics. This definition does not require acyclicity. A particular runtime may accept only a strict DAG, while another may admit cycles, event-triggered transitions, or bounded iteration.

Mechanism family	Primary or official anchors	Established precedent or paper role
Design-space decomposition and lowering	Build Systems à la Carte; Pegasus; CWL [?, ?, ?]	Factorized design coordinates and a distinction between abstract and resource-mapped or portable executable form
Define-by-run and traced graph extraction	TensorFlow Eager [?]	Imperative construction, graph extraction, and staged execution within one system are direct positive precedents; addressability and admission remain separate discriminants
Stateful graph runtime and persistence	LangGraph runtime, persistence, and compatibility documentation [?, ?, ?]	Graph execution, checkpointed state, and explicit compatibility policy are established operational mechanisms
Planning, stored plans, and adaptive construction	PLANEX; LLMCompiler; Xu et al.’s execution-time EvoMAS [?, ?, ?]	Stored-plan execution, planner–executor separation, and execution-time topology adaptation are established mechanisms
Content identity and replay	Nix; Bazel REAPI; reproducible builds; Durable Functions; Temporal [?, ?, ?, ?, ?]	Declared-input identity, bit-reproducibility, and deterministic event replay are distinct precedent classes
Pre-effect policy and authority	PCC; Kubernetes; SLSA; Sigstore policy-controller; Gatekeeper; AgentSpec; SARC [?, ?, ?, ?, ?, ?, ?]	Pre-execution checking, provenance verification, admission hooks, policy-as-code, and agent action governance precede the narrower artifact-admission construct
Provenance and accountable evidence	in-toto; Sigstore; PROV-DM [?, ?, ?]	Authenticated supply-chain history and standardized provenance vocabularies precede the paper’s evidence layer
Author-checked Mentu instantiation	Current committed-source census and Core tests (Section 6)	Instantiates the paper’s strict-DAG admitted profile; it is not evidence of priority or comparative rank

Table 1: Positive-only precedent map. Each scholarly row states a mechanism that its cited source establishes. No blank cell or omitted system supports an absence, uniqueness, or novelty inference.

Any operation that creates G , adds or removes a member of V or E , or mutates an effect-bearing value in λ , κ , or σ_g is construction-lane activity. Selecting a ready node under an unchanged, already admitted σ_g is execution scheduling, not graph construction.

Lowering combines the graph with an effective execution policy P to produce an effect-semantic executable payload

$$\chi = (V, E, \lambda, \kappa, \sigma_g, P),$$

where P contains only runner-consumed choices fixed in the executable, such as a frozen back-end/model selector, limits, and other executable-bound policy. Granted permissions, credentials, sandbox/approval posture, and admission-resolved routing are not moved into P . A separate artifact header

$$\nu(x) = (\nu_L(x), \nu_C(x), \nu_\beta(x))$$

names the lowerer, canonicalizer, and binding-schema versions. The executable artifact is $x = (\nu(x), \chi)$. This split is intentional: a semantics-preserving toolchain bump need not be called graph-semantic, yet the artifact can still bind the interpretation contract under which its bytes were produced.

A *workflow* describes process intent or control semantics and may be authored in a domain-specific notation or program. It is not automatically an agent graph. A *plan* is an addressable lifecycle artifact that records the candidate, its lowering and qualification context, and the executable identity needed for later requalification. A *state machine* specifies states and transition rules; it may implement scheduling or be compiled into a graph. An *executable intermediate representation* (IR) is the normalized form consumed by the runner. An *authoring method* changes how a source is produced; it grants no runtime authority by itself. “Profile” is reserved below for a complete runtime compatibility surface, while ψ names an assurance contract.

Let o be an objective, s observed repository and configuration state, a an external authority envelope, g a candidate graph, x canonical executable IR, q a qualification report, z the frozen plan container, c a concrete admission-time runtime context, η the bindable residual environment, ξ exogenous state not bindable at admission, r an admission receipt, y an execution trace and output, and e the resulting evidence. For an externally pinned toolchain and schema profile $\bar{\nu} = (\bar{\nu}_L, \bar{\nu}_C, \bar{\nu}_\beta)$, the maximal staged lifecycle is typed as

$$\begin{aligned} g &= \text{Construct}_\pi(o, s_0, a), \\ x &= \text{Lower}_{\bar{\nu}}(g, P), \\ q &= \text{Qualify}_{\bar{\nu}}(x, a, s_0), \\ z &= \text{Freeze}(x, q, s_0), \\ (x_{\text{rq}}, q_{\text{rq}}, c_1) &= \text{Requalify}_{\bar{\nu}}(z, a, s_1), \\ r &= \text{Admit}(x_{\text{rq}}, q_{\text{rq}}, a, c_1), \\ y &= \text{Run}_\rho(x_{\text{rq}}, r), \quad e = \text{Seal}(y, r). \end{aligned}$$

The tuple accepted by each operator is now explicit; the notation does not pretend that freezing consumes a qualification report while losing the executable it must preserve. *Mechanical qualification* here means the declared schema, dependency, contract, and budget checks performed by the runtime. It is not equipment, process, or tool qualification in the regulated engineering sense and does not certify fitness for a regulated use. The residual environment η is projected only after the qualification and concrete runtime context it binds are available; it is not an input to initial qualification. Not every policy exposes every stage as a separate invocation. Dynamic one-shot

construction may pass from qualification to admission without an operator-visible artifact. Static execution begins with a source frozen before the measured invocation. The model’s claim is that those policy differences need not change ρ .

Symbol	Meaning
χ	Effect-semantic executable payload $(V, E, \lambda, \kappa, \sigma_g, P)$
$\nu, \bar{\nu}$	Artifact header and externally selected compatible lowerer/canonicalizer/binding-schema profile
β, D_β, p_β	Declared field inventory, placement map, and field-equivalence relations
F_ν, F_χ, F_η	Header, executable-payload, and residual-environment field partitions
$x_{\text{rq}}, x_{\text{succ}}$	Requalified executable and a semantically mutated successor requiring a new identity
μ, ψ, ρ	Authoring method, assurance contract, and execution substrate

Table 2: Core notation. Symbols not listed here are introduced locally in the lifecycle definition.

Admission is non-mutating and identity-preserving: it either refuses or emits a receipt binding the unchanged executable. An admission mechanism that changes an effect-semantic executable field has produced a successor x_{succ} , which must be requalified and admitted under its own identity before effectful dispatch. Mutation cannot be hidden inside the authority transition in this model.

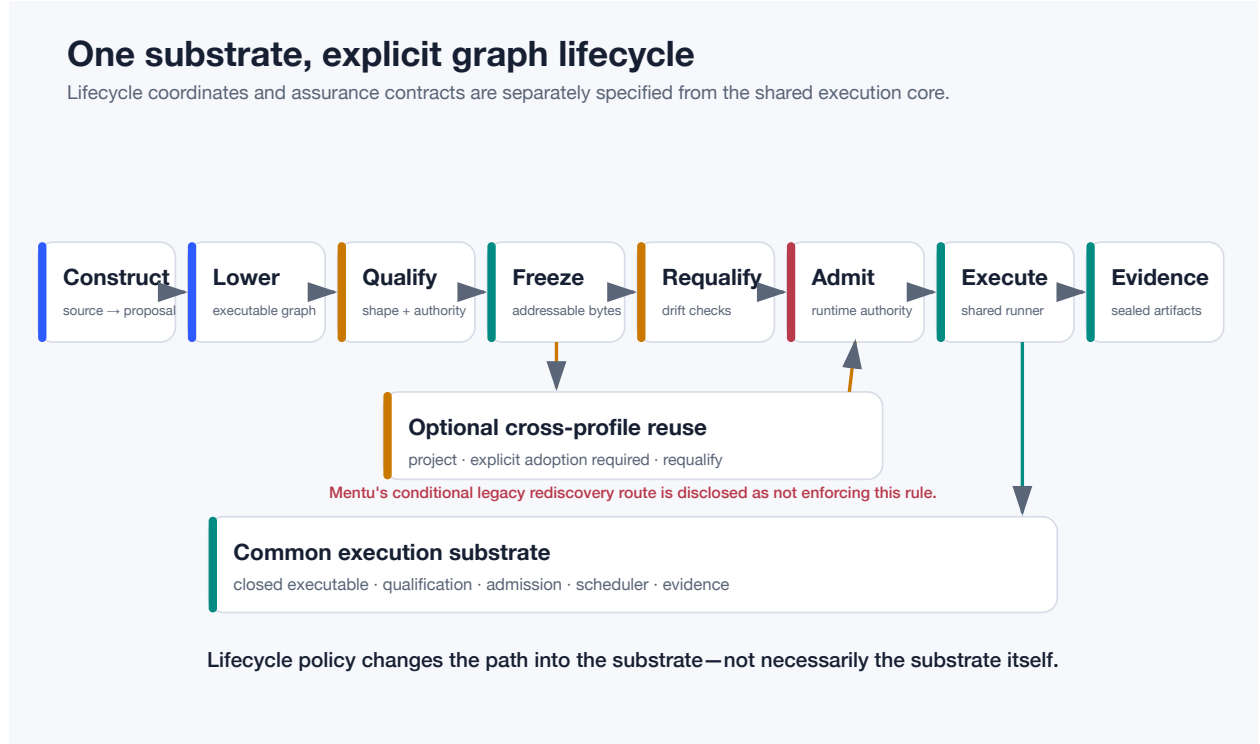


Figure 1: The maximal graph lifecycle. Policy changes occur before execution; the common substrate executes the qualified graph, while authority-entry convergence is evaluated separately.

Binding schema, artifact header, and non-equivalent identities

Let the binding schema selected by the external pin be

$$\beta_{\bar{\nu}_\beta} = (D_\beta, p_\beta, \{\approx_f\}_{f \in D_\beta}, \text{type}).$$

Here D_β inventories declared bindable fields independently of the artifact and receipt encoders, while

$$p_\beta : D_\beta \rightarrow \{\nu, \chi, \eta\}, \quad F_k^\beta = p_\beta^{-1}(k), \quad D_\beta = F_\nu^\beta \dot{\cup} F_\chi^\beta \dot{\cup} F_\eta^\beta.$$

The exact-match identity-header class is $F_\nu^\beta = \{\nu_L, \nu_C, \nu_\beta\}$; F_χ^β contains payload fields and F_η^β contains residual environment fields bound at admission. Superscripts are omitted when β is fixed. Field types and per-field equivalence relations are declared independently of the canonicalizer. The mandatory payload core is

$$\{V, E, \lambda, \kappa, \sigma_g, P\} \subseteq F_\chi.$$

Placement uses separate substitution domains. Intrinsic executable semantics

$$\mathcal{I}_\rho(\chi) = (\mathcal{L}_\rho(\chi), K_\rho^\chi)$$

contains the unfiltered language of control, proposal, verification, and attempted-effect traces and any executable-bound transition law. Holding ρ fixed, a lowering-time field whose well-typed one-field payload substitution changes $\mathcal{I}_\rho$ belongs in F_χ . Node identity, behavior, contracts, precedence, graph-declared scheduling, and executable-bound policy are mandatory instances.

Authority-conditioned semantics is a different object:

$$\mathcal{A}_\rho(\chi, \eta) = (\text{Admissible}_\rho(\chi, \eta), \mathcal{L}_\rho^{\text{allowed}}(\chi, \eta), K_\rho^{\text{run}}(\chi, \eta)).$$

Holding (ρ, χ) fixed, an admission-time field whose well-typed one-field environment substitution changes that decision or allowed trace projection, or the context-resolved execution law, belongs in F_η ; it does not redefine $\mathcal{I}_\rho(\chi)$. Requested effect scope is a payload contract. Granted permission, credentials, sandbox/approval posture, and independently resolved routes are residual environment facts. A model/backend selector is F_χ only when frozen in the executable and thus part of K_ρ^χ ; a route resolved independently at admission is a distinct F_η fact. Provider sampling after admission is ξ . Lowerer, canonicalizer, and schema versions remain F_ν provenance. An omitted, bindable fact cannot be relabeled exogenous to protect the schema.

These placement predicates are denotational; the paper does not assume that equality of arbitrary stochastic trace languages is decidable. Operational tests instead declare a finite observation profile

$$\omega = (h, \pi_\omega, \zeta),$$

where h is a transition horizon, π_ω is an event projection (subscripted to keep it distinct from the lifecycle policy vector π ; finding BR9-004 of the v0.9.9-r6 blind review), and ζ is a reproducible choice harness. A controlled substitution that changes the resulting observation is a bounded witness against equivalence. An unchanged finite observation is only non-falsification under that ω ; it does not prove denotational equality or causal inertness.

The v0.9.9-r6 blind review observed (finding BR9-001) that a profile left to the author and absent from the published evidence vector makes non-falsification unearned: a projection too coarse to

distinguish anything yields it for free. The concrete profile is therefore published. Every non-falsification this paper reports was observed under one profile, ω_0 : the horizon h is the complete archived run record of each reported run, from admission through terminal state, untruncated; the event projection π_ω is the canonicalizer's identity-bearing field projection at the pinned F_ν versions; and the choice harness ζ is the committed reproduction wrapper at its archived commit, executed against byte-fixed archives with no live provider sampling, so ξ is not exercised. ω_0 is carried as the eighth component of Γ_E , and every reported non-falsification is paired with the positive detection controls of Section 8 — three of three historical disposable-copy mutations detected, one of one current-Core admission-drift control detected — so a profile that detects nothing is distinguishable from a genuine equivalence: under ω_0 the detectors fire on seeded differences and do not fire on the reported substitutions. An adequacy condition constraining π_ω in general remains open and unclaimed.

Effect-semantic equivalence is explicit:

$$\chi_1 \equiv_\beta \chi_2 \iff \forall f \in F_\chi, \text{value}_{\chi_1}(f) \approx_f \text{value}_{\chi_2}(f).$$

The per-field relations also carry a trace-soundness obligation for every substrate contract ρ claimed by the profile:

$$\text{TraceSound}_\rho(\beta) \iff \forall \chi_1, \chi_2, \quad \chi_1 \equiv_\beta \chi_2 \Rightarrow \mathcal{I}_\rho(\chi_1) \equiv \mathcal{I}_\rho(\chi_2).$$

Finite observations can falsify this obligation but cannot certify it universally. This keeps a coarse field-equivalence rule from silently erasing a trace-distinguishing payload change while leaving byte canonicalization formally conforming to that coarse rule. Residual-environment equivalence is defined independently:

$$\eta_1 \equiv_{\eta, \beta} \eta_2 \iff \forall f \in F_\eta, \text{value}_{\eta_1}(f) \approx_f \text{value}_{\eta_2}(f).$$

It carries the parallel environment/authority obligation

$$\text{AuthoritySound}_{\rho, \chi}(\beta) \iff \forall \eta_1, \eta_2, \quad \eta_1 \equiv_{\eta, \beta} \eta_2 \Rightarrow \mathcal{A}_\rho(\chi, \eta_1) = \mathcal{A}_\rho(\chi, \eta_2).$$

A coarse relation may therefore not erase a change in admission, allowed traces, or the context-resolved execution law. A controlled $\text{Obs}_{\rho, \omega}^A$ difference falsifies authority soundness under the declared profile; finite equality is only non-falsification, never a universal certificate. Environment canonicalization must preserve the declared relation rather than silently substitute pathname or serialization equality.

The accepted identity profile $\bar{\nu}$ is selected outside the artifact; the header does not recursively select the function that interprets it. Before canonicalization or qualification, the strict model checks

$$\text{HeaderOK}_{\bar{\nu}}(x) \iff \nu(x) = \bar{\nu}.$$

A mismatch refuses before canonicalization and before host effects. Only then does the externally selected canonicalizer consume χ . It conforms to $\beta_{\bar{\nu}, \beta}$ only when both directions hold:

$$\text{canon}_{\bar{\nu}_C, \beta}(\chi_1) = \text{canon}_{\bar{\nu}_C, \beta}(\chi_2) \iff \chi_1 \equiv_\beta \chi_2.$$

Canonical artifact bytes separately frame a fixed header encoding and the canonical payload:

$$b_{\bar{\nu}, \beta}(x) = \text{frame}(\text{canonHeader}(\nu(x)), \text{canon}_{\bar{\nu}_C, \beta}(\chi)), \quad \text{HeaderOK}_{\bar{\nu}}(x).$$

Thus a header change creates a new artifact identity without being mislabeled as a trace-semantic mutation. In the MENTU instantiations and paper tools, H is full-length SHA-256 and executable identity is

$$h_x = \text{SHA256}(b_{\bar{\nu},\beta}(x)).$$

For run j , define the residual binding after requalification and concrete context selection as

$$\eta_{\beta,j} := \text{bind}_{F_\eta^\beta}(s_j, a_j, q_{\text{rq},j}, c_j).$$

Environment identity is the separate binding $h_{\eta,j} = \text{SHA256}(\text{canon}_{\eta,\beta}(\eta_{\beta,j}))$, when an aggregate is computed. Truncated hexadecimal strings in figures are labels, never identities.

Two tests must remain separate. For admitted run j , let U_j inventory identity-bearing field names actually recovered from the executable and receipt, and let $O_j \subseteq U_j \times \{\nu, \chi, \eta\}$ record their observed placement without copying p_β . *Declared binding closure* requires

$$U_j = D_\beta, \quad \forall f \in D_\beta, \quad \{k : (f, k) \in O_j\} = \{p_\beta(f)\}.$$

This compares an independent declared inventory with observed encoding and requires complete, unique placement; it is not merely the declared partition identity above. *Schema adequacy* asks whether a bindable causal fact was omitted. An omission probe holds x and declared η fixed, varies one undeclared candidate fact such as execution-model identity, permission posture, clock, or locale, and observes whether admission, the allowed trace projection, or the context-resolved execution law changes. A positive witness is a schema defect and bars an adequacy claim under that β .

Before admission, the schema itself must also be well formed: p_β is total and typed on D_β ; its three inverse images are pairwise disjoint; every non-header identity field has exactly one declared semantic-class assignment with an addressable analytic or external justification; and the independently enumerated identity-bearing names emitted by executable and receipt encoders equal D_β . The mechanical gate checks declaration presence, uniqueness, placement, and encoder closure; it is not a semantic oracle. Unassigned names, names without a declared semantic class, dual-domain, undeclared-emitted, declared-but-unemitted, and multiply placed names cause refusal. Trace soundness, authority soundness, and open-world adequacy remain semantic obligations that finite witnesses can falsify but cannot prove.

RFC 8785 specifies a canonical JSON serialization for stable hashing, including primitive serialization and deterministic property ordering [?]. The model's canonicalizer must state whether it conforms to that baseline or deliberately differs. Its declared semantic equivalence, path/content aliases, field placement, and artifact framing are domain extensions; RFC 8785 does not establish them. Canonicalizer evidence likewise needs both directions: well-typed semantic mutations must produce distinct canonical bytes, while equivalent encodings such as key ordering, normalized numbers, and declared path equivalents must converge. An F_η -only change must leave h_x unchanged while changing the environment binding or admission decision. The present evidence does not establish complete schema adequacy or full biconditional canonicalizer conformance. Neither the historical C30 pin nor the current Core pin exercises a HeaderOK $_{\bar{\nu}}$ mismatch-before-effects fixture, so header refusal remains a specification obligation rather than an observed pass at either pin.

In the historical C30 MENTU path at 40bd3f9, the full **SequenceDefinition** binds nodes and **dependsOn** relations, while the frozen backend/model selector, other executable-bound policy, and lowerer version are also bound by the executable artifact. Under the refined model, the graph and executable-bound policy fields belong to F_χ , while the lowerer version belongs to F_ν . Repository

and discovery state, granted authority, qualification state, resolved credential/endpoint posture, and concrete admission context remain in F_η . This field-placement example interprets the historical pilot; it is not projected forward onto the current Core definition.

Even if both identities match, an agent node may be stochastic. Output reproducibility is therefore a property of observations from

$$Y \sim \mathcal{D}(\cdot \mid x, \eta, \xi),$$

where ξ includes model sampling, remote service behavior, and other uncontrolled state. Semantic judgment has two distinct targets: $\text{PlanAdequate}(x, o)$ asks whether the executable expresses an appropriate course of action for the objective, while $\text{OutcomeCorrect}(y, o)$ asks whether an observed outcome satisfies it. Graph identity, environment identity, output reproducibility, plan adequacy, and outcome correctness require different evidence and are never substituted for one another in this paper.

Projection is not adoption

A frozen executable may be projected into a persistent source form p for evidence or reuse. Projection parity requires that pinned lowering reproduce the executable identity bound by z ; it does not grant authority. If p later enters a different assurance contract ψ_t , the transition is explicit:

$$\text{Project}(z) \rightarrow p, \quad \text{Adopt}(p, \psi_t, a, s) \rightarrow (x_t, q_t) \mid \text{refuse}.$$

Adoption names the target contract, re-lowers or loads the artifact, qualifies it, and requires that contract’s admission transition before effects. Discoverability through another parser or command is not adoption. This rule keeps authoring origin out of runtime authority while preventing parseable bytes from silently inheriting the weakest available entry path.

4 Lifecycle policies

The C30 policy vector is

$$\pi = (\tau, \phi, \mu, \rho),$$

where τ is construction time, ϕ is the freeze boundary, μ is authoring method, and ρ is execution substrate. Let

$$\Omega \subseteq T \times \Phi \times \mathcal{M} \times R$$

be the feasible policy region. The coordinates are separately specified: changing one does not redefine the others. This is a separability claim over Ω , not a claim that every Cartesian combination is feasible, independent in a statistical sense, or free of engineering constraints. The named modes are reference configurations, not an exhaustive taxonomy. The normative coordinate vocabulary is: $\tau \in \{\text{pre-task}, \text{task-time}\}$; $\phi \in \{\text{continuous}, \text{addressable}\}$; μ names the source-production method; and ρ names the runner-and-scheduler substrate whose execution semantics are being compared. The plan container z is the addressable lifecycle record used by addressable- ϕ policies; z binds x but is not synonymous with it.

An entry policy also has an assurance contract $\psi_p = (\mathcal{Q}_p, \mathcal{A}_p, \mathcal{E}_p)$, comprising its qualification regime, admission transition, and evidence contract. This is not a fifth C30 treatment axis. It makes explicit that $\rho_p = \rho_q$ does not imply $\psi_p = \psi_q$: policies can share execution semantics while

retaining different assurance contracts. Here ψ is an author-declared runtime contract, not an independently evaluated security profile, protection profile, or certification level.

Mode	Construction time	Freeze boundary	Authoring method	Execution substrate
S	Before task invocation	Addressable source before invocation	Method μ_S	Shared graph substrate (model)
D	At task invocation	Continuous construction-to-execution invocation	Method μ_D	Shared graph substrate (model)
H-direct	At task invocation	Addressable plan before a later execution invocation	Direct planner	Shared graph substrate (model)
H-scaffold	At task invocation	Addressable plan before a later execution invocation	Scaffold authoring method	Shared graph substrate (model)

Table 3: Reference policies as configurations of separately specified lifecycle coordinates in the feasible region Ω . “Shared” is a model parameter and an implementation question, not a universal fact about all systems.

These four points do not empirically demonstrate independent variation of all coordinates: ρ is held constant, and the observed τ, ϕ values are only partially crossed. Separability is therefore a specification claim about how a configuration is described, not an identified causal effect or a claim that the reference points span Ω .

Two consequences follow from the factorization. First, “scaffold” is not a runtime kind. Here it names an authoring method, not the broader model-and-tool harness also called a scaffold in agent literature. If it changes only μ , then a scaffolded candidate must satisfy the same lowering, qualification, admission, and execution contracts as a directly authored candidate. Second, a static source need not have been written by a human, and a dynamically constructed source need not execute immediately. Provenance, timing, freezing, and execution are separable.

The word *hybrid* remains useful only when its boundary is named. Here it means dynamic construction followed by an explicit, addressable freeze and a later requalification-and-execution invocation. It does not mean two runtimes, weaker contracts, or continued planner authority while the frozen epoch runs.

Four separate evaluation decisions

The model does not use one word, “implemented,” for four different questions. First, an *author-checked architectural instantiation* requires an origin-neutral executable type, one declared scheduler contract, and mapped construction or loading boundaries that can target them. Committed source and named engineering tests must support that mapping. This label denotes an author-performed check, not an independent audit or canonical reference status.

Second, *executed cross-origin parity* is stronger: at least two feasible lifecycle policies must lower or load equivalent fixtures, enter the same runner, receive the same node-dispatch and graph-declared scheduling semantics, and show that origin-specific metadata does not silently alter those semantics. The evaluation must follow and execute both traces; a source definition or unexecuted fixture is insufficient.

Static, dynamic, and hybrid as lifecycle policies

The policy coordinates are separable over a feasible region; the arms are configurations, not four engines.

Arm	Construction	Freeze	Authoring	Execution
S	before task	persistent	recorded prior	shared substrate
D	task time	continuous	protocol-fixed	shared substrate
H-direct	task time	staged	direct	shared substrate
H-scaffold	task time	staged	scaffold	shared substrate

Scaffold is an authoring method; staging is a freeze policy.

Figure 2: S, D, H-direct, and H-scaffold occupy points in one policy space. The labels do not define separate runner implementations.

Third, full staged binding/admission conformance requires I1–I6 under fixed versions, including exercised I4 drift categories and complete I5 declared-closure and mandatory-field mutation fixtures. Shared machinery alone is insufficient.

Fourth, logical safety is reported separately from operational evidence adequacy. Implications can pass vacuously in a runtime that refuses everything. We therefore report

$$\Gamma_E = (n_{\text{recorded host-boundary runs}}, n_{\text{distinct } x}, n_{\text{projection}}, |J_{\bar{\nu}, \beta, f}|, n_{\text{header refusal}}, \mathcal{D}_{I4}, \mathcal{M}_{I5}, \omega).$$

The eighth component publishes the concrete observation profile under which the reported non-falsifications were observed (Section 3, ω_0), so the strength of every equivalence statement is inspectable rather than implicit. Γ_E is published for the historical population only, by design at this candidate: current-Core engineering tests are reported outside this vector, and any future current-pin claim must carry its own Γ_E (finding BR9-005). At least one admitted run with recorded host-boundary activity is required for instrumented non-vacuity; this count is not proof that every semantic effect was observed. A cross-input identity claim requires at least two executable identities; I3 requires a projection; I1 must expose its eligible population; and every claimed header, I4, or I5 category requires a seeded or observed witness. The historical pilot is non-vacuous for I1–I3 but has no header-refusal fixture, I4 challenge, or complete I5 matrix. Paper-local detector controls and current Core tests are separate evidence, not retroactive pilot observations.

5 Qualification, freeze, admission, and execution invariants

The lifecycle becomes operational only when its transitions have testable obligations. The author check of the pinned commit supports an architectural instantiation for the named strict-DAG Core profile. Current tests exercise origin-neutral lowering, an admitted vertical slice, saved-plan execution, canonicalization, and drift refusal, but they do not execute an ordinary static recipe and a generated graph through one Core admission route. Product-wide executed cross-origin parity is therefore not claimed. Full staged-protocol conformance additionally requires I1–I6 under fixed versions, exercised I4 drift classes, and a complete I5 fixture matrix. C30 tests only a subset of that stronger protocol at its separate historical pin.

I1—staged executable identity. Fix $\bar{\nu}$, $\beta = \beta_{\bar{\nu},\beta}$, and receipt field $f = \text{executableId}$. Let $J_{\bar{\nu},\beta,f}$ contain the staged invocations for which requalification succeeds, no bound-field drift is detected, and admission returns a pre-effect receipt exposing f . For every $j \in J_{\bar{\nu},\beta,f}$:

$$\text{SHA256}(b_{\bar{\nu},\beta}(x_{\text{inspect},j})) = \text{SHA256}(b_{\bar{\nu},\beta}(x_{\text{rq},j})) = r_j \cdot f.$$

If the saved artifact stores a candidate rather than only an executable, re-lowering must reproduce the saved executable bytes under the pinned lowerer and canonicalizer. The pilot instantiates this condition only for the MENTU v1 lowerer, canonicalizer, commit, and receipt field named in the author source check. The identity ratio is conditional on reaching J , so every report must also publish the attempted, requalification-refused, admission-refused, admitted, and identity-matched counts. It is not a pipeline success rate.

I2—no execution-time graph planning. Let J_{saved} contain invocations registered before any run telemetry as entering through the saved-plan requalify-and-admit route with an existing addressable plan. Membership is fixed by that entry route, not by the later dispatch outcome. The invariant is

$$\forall j \in J_{\text{saved}}, \quad N_{\text{Construct},j}^{\text{exec}} = 0.$$

A model call that implements an admitted node is execution, not graph planning; the two lanes must be separately classified. An observed graph-construction dispatch is an I2 failure retained in J_{saved} , not grounds for post-outcome reclassification.

I3—persistent projection parity. When a generated executable is materialized as a persistent recipe projection, their canonical executable identities must be equal. This establishes parity for the materialized artifact; it does not move all persistent recipes through the generated path or prove global admission convergence. A parity pass is the conjunction of lossless projection and deterministic pinned lowering; without a separate lowerer-determinism control it does not identify which component would explain a failure.

I4—drift refusal before effect. Executable identity does not waive current-state checks. Requalification must bind the frozen graph to the current η_1 , and drift outside the admitted policy must refuse before host dispatch and target-path change. Requalification may reject exact bytes; rejection preserves, rather than contradicts, I1. C30 prospectively specifies this invariant, but the pilot deliberately challenged zero drift classes. Its source mapping and prospective specification are not empirical drift-test evidence.

I5—declared binding closure. For every admitted run j , the independently declared inventory and placement map must agree with the observed executable/receipt inventory and placement relation:

$$U_j = D_\beta, \quad \forall f \in D_\beta, \quad \{k : (f, k) \in O_j\} = \{p_\beta(f)\}.$$

The executable binds every field assigned to F_ν and F_χ ; the receipt binds every declared F_η constituent or an aggregate identity with declared equality semantics, a schema-versioned ordered constituent inventory, and addressable evidence from which the aggregate can be recomputed. An opaque aggregate cannot satisfy $U_j = D_\beta$. An extra, missing, duplicated, or misplaced identity-bearing field, a header/pin mismatch, or an unbound declared environmental constituent causes refusal before effect. A failed payload inequivalence, equivalence, partition-isolation, or header fixture makes the relevant binding or canonicalization claim nonconforming under that β .

The C30 pilot observes selected executable and receipt bindings, but it does not compare an independent complete D_β inventory with U_j , and it does not exercise a complete mandatory-field mutation suite. It therefore does not establish full declared closure or biconditional canonicalizer conformance. Neither the C30 pin nor the current Core pin exercises HeaderOK _{$\bar{\nu}$} mismatch refusal. Schema adequacy is a separate open-world property requiring omission probes over plausible undeclared bindable facts; neither declared closure nor a conforming canonicalizer establishes it.

D1—authoring/authority non-implication. A candidate, qualified candidate, and frozen plan have no implied effect authority. Admission produces a bounded authority state, and changing μ cannot change that state when the executable, qualification result, external authority envelope, and bound runtime context are equal. If an explicit qualification, authority, or runtime contract names the profile, the authoring method becomes an authority-bearing policy input and the implication no longer applies. This is a construct boundary, not an empirical invariant: scaffold describes construction unless a named authority contract says otherwise.

I6—evidence integrity. For this model the run record must bind six things: the executable, authority envelope, qualification result, concrete runtime context, run identity, and artifact manifest. This is a closed mandatory set. Every manifest file digest and its derived aggregate bundle digest must recompute. I5 establishes field placement; I6 establishes that the receipt, run identity, and artifact manifest survive as addressable evidence. The pilot verified receipt fields and manifest digests but did not compute a standalone aggregate h_η . At the current MENTU pin, the concrete runtime context is not an unnamed placeholder: it includes the run and workspace roots, transfer mode, locks and baselines, Git and selected-read-set digests, dependencies and variables, the resolved backend/model route (distinct from any executable-frozen selector), auth, endpoint, permission, sandbox, approval, MCP, primitive and adapter identities, engine version, qualification/runtime state digests, and outer-call/gate posture. The executed drift control perturbs the declared `runtimeStateDigest`; it is an F_η -side admission check, not an executable-field mutation. Evidence is a product of execution, not a transcript substituted for an addressable artifact.

D2—semantic non-implication. Mechanical qualification is sound only relative to its declared checks. Passing schema, dependency, footprint, budget, or verification predicates does not prove plan adequacy or outcome correctness. In particular:

$$\bigwedge_{k=1}^6 I_k \not\Rightarrow \text{PlanAdequate}(x, o), \quad \bigwedge_{k=1}^6 I_k \not\Rightarrow \text{OutcomeCorrect}(y, o).$$

D1 is excluded because it is an analytic construct boundary, not an empirical protocol invariant.

Scope theorem: runtime replanning

Let receipt r_0 admit x_0 , whose graph-semantic core is G_0 . After execution begins, suppose a planner changes a node, dependency, behavior binding, node contract, or graph-declared scheduling transition to produce x_{succ} . The narrow identity lemma has the explicit antecedent

$$\chi_{\text{succ}} \not\equiv_{\beta} \chi_0.$$

Given that antecedent and a canonicalizer conforming to β :

$$b_{\bar{\nu},\beta}(x_0) \neq b_{\bar{\nu},\beta}(x_{\text{succ}}).$$

Under the stated collision-resistance assumption,

$$\text{SHA256}(b(x_0)) \neq \text{SHA256}(b(x_{\text{succ}})) \implies r_0 \text{ does not bind } x_{\text{succ}}$$

except in the digest-collision case. Dispatching x_{succ} under r_0 violates the non-mutating admission binding. A same-byte trace-relevant mutation exposes a trace-unsound β , a field-placement defect, or canonicalizer nonconformance; it is not an identity distinction. A distinct-byte same-digest observation challenges the digest assumption. A scheduler-build change rather than graph-declared scheduling changes the bound environment identity or causes refusal.

The stronger semantic corollary may derive the narrow antecedent only when

$$\mathcal{I}_{\rho}(\chi_{\text{succ}}) \not\equiv \mathcal{I}_{\rho}(\chi_0) \quad \wedge \quad \text{TraceSound}_{\rho}(\beta).$$

A coarse relation that treats $A \rightarrow B$ and $B \rightarrow A$ as equivalent because they share endpoint names can preserve canonical bytes while reversing observable order. That is a counterexample to trace soundness of the schema, not evidence that identity distinguished the mutation. A finite $\text{Obs}_{\rho,\omega}^I$ difference is a bounded witness, not a universal certificate.

An *admission-scoped graph epoch* is the interval governed by one receipt and executable identity. The term borrows the fencing intuition of a generation change but makes no claim about distributed consensus, leader election, or a product-wide epoch counter.

There are two coherent alternatives. The system can close epoch e_0 , construct and qualify G_1 , and issue a new admission for epoch e_1 ; or it can define an adaptive runtime in which the planner and its allowed transition relation are themselves part of the admitted executable semantics. The latter may keep a fixed outer graph while producing data-dependent work, but it is not execution of the same frozen inner topology. Runtime replanning that changes graph structure or scheduling semantics without either treatment is outside the frozen-epoch protocol population. An interpreted surface that claims neither a complete graph nor an admitted meta-runtime identity is a third honest classification. This is a scope boundary, not an empirical counterexample or falsifier of the model. Its value is discriminant: it states which object must carry authority when topology remains adaptive. A fixed outer behavior tree is one established way to represent data-dependent control without silently changing the admitted inner object [?].

6 Mentu as an Author-Checked Architectural Instantiation

The evidence is intentionally split across two source epochs. Historical C30 observations and their original source corroboration remain pinned to 40bd3f9. The current architecture mapping is pinned independently to f3d89e7. The later Core extraction is not inferred backward into the pilot.

At the current epoch, MENTU is the paper’s author-checked architectural instantiation of the strict-DAG admitted Agent Graph Runtime profile. “Author-checked” means an author-proposed model-to-system mapping checked against committed source and named engineering tests executed by the author-side reproduction harness. It is not an independent audit, external certification, canonical reference status, a product-wide uniformity claim, or a comparative rank.

An extracted admitted graph Core with a public API. `MentuExecutionGraphCore` owns the closed `ExecutionGraphDefinition`, lowering, canonicalization, qualification, admission, activation evidence, lifecycle types, and `ExecutionGraphScheduler`. The checked `MentuSequenceDefinitionAdapter` is the boundary from `MentuEngine`’s general recipe schema to the admitted-only Core definition. `MentuEngine` retains host effects, product integration, and presentation through adapters. Core rejects unknown dependencies, self-dependencies, and cycles, so the admitted profile is a strict DAG; neither the abstract model nor every Mentu surface is therefore DAG-only.

One admitted scheduler, explicit compatibility profiles. Equivalent admitted definitions receive DAG validation, ready-set computation, bounded parallelism, exclusivity, failure cascade, and circuit behavior from the one Core scheduler. The engine’s `StepScheduler` projects engine state and bridges effectful node execution rather than defining a second DAG scheduler. The prepared admitted path constructs the sole admitted `SequenceRunner` only after Core admission and pre-host state validation. Committed current-head test definitions cover persistent/generated origin parity, admission drift refusal, and an admitted vertical slice. The paper-local current-epoch verifier executes 13 named filters from an exact Git archive with external build scratch and an isolated runtime home: 25 of 25 targeted tests pass. Coverage includes Core and engine canonicalization, generated/persistent lowering parity, admission identity, the admitted vertical slice, saved-plan exact execution, projection parity, lifecycle ordering, pre-runner drift refusal, and the Engine–Core round trip. These are author-side engineering tests, not independent black-box replication or exhaustive schema conformance. None of the 13 filters exercises `HeaderOKp` mismatch refusal.

The current-epoch mapping follows generated and saved-plan prepared boundaries to the same Core definition and scheduler, while its origin-parity fixture compares equivalent generated and persistent lowerings. Historical C30 runner evidence remains corroboration for its own pin and is not substituted for this current-Core mapping. No test in this revision executes an ordinary static recipe and a generated graph through one Core admission route, so product-wide executed cross-origin parity remains unclaimed.

Mentu also retains deliberate compatibility profiles. Ordinary recipe dispatch can use the raw `SequenceRunner` initializer with legacy authority, and `mentu workflow run` uses the interpreted `WorkflowRunner`. Their different assurance or execution contracts are profile boundaries, not evidence that the admitted graph core is architecturally incomplete. They are also not silently counted

Recipe auto	CLI auto	Effective auto	Commit-record gate	Authority entry
false	false	false	no	legacy
false	true	true	no	legacy
true	false	true	yes	legacy after gate
true	true	true	no	legacy

Table 4: Committed-source control-flow matrix for ordinary recipe dispatch. The gate is a recipe-commit record existence check, not Core qualification or admission. The matrix reports no operational traffic.

as conformance evidence for that Core.

The current committed-source census found 11 persisted recipe type cases, 10 direct production **SequenceRunner** construction sites (9 raw and 1 admitted), and 2 direct **WorkflowRunner** construction sites. These are exact lexical constructor counts, not a denominator or percentage of every effectful Mentu entry point. The audit classifies the graph-bearing roots it names and leaves other product operations outside its population. A lexical census establishes where constructors are written, not which are reachable or how often they execute: the sole-admitted-runner statement it supports is a source-layout claim, and a reachability-grade check (call-graph or runtime instrumentation over the census population) is registered as future work, not claimed (finding BR9-006 of the v0.9.9-r6 blind review).

Current projection-adoption boundary. The source audit found one concrete cross-profile transition boundary. Persistent projection writes a normal `recipe.json`, prompts, and parity evidence, but no required-authority marker. If placed under ordinary recipe discovery, those files can later be parsed by `mentu run` and reach legacy compatibility authority without an explicit adoption transition. Recipe dispatch computes effective auto behavior as CLI auto OR recipe auto. It nevertheless constructs the raw **SequenceRunner** after any applicable precondition in all four Boolean cells, and that initializer assigns legacy authority independently of the effective-auto value:

The unmodified generated projection occupies the recipe-auto-false row because materialization adds no `auto_mode` marker; CLI absence and explicit false collapse to its false cell. The generated graph invocation itself enters through the admitted route. The coherent alternate-contract transition is:

$$\text{generated projection} \rightarrow \text{Adopt}(\psi_t) \rightarrow \text{target qualification/admission} \rightarrow \text{execute}.$$

Until Mentu enforces that transition, admitted authority must be described as a property of the qualified-and-admitted invocation path, not an intrinsic property of parseable projection bytes. This is a bounded cross-profile adoption finding exposed by the model, not a verdict that the runtime abstraction is missing. The source matrix establishes reachability and control flow, not successful executions or use frequency in any cell.

MENTU makes the abstraction operational and inspectable: exploratory construction can converge on a closed executable whose qualification, admission, scheduling, activation evidence, and host boundary are explicit. The current source establishes an admitted reference profile and named compatibility surfaces. It does not establish global adoption enforcement, exhaustive I5 mutation coverage, or comparative superiority.

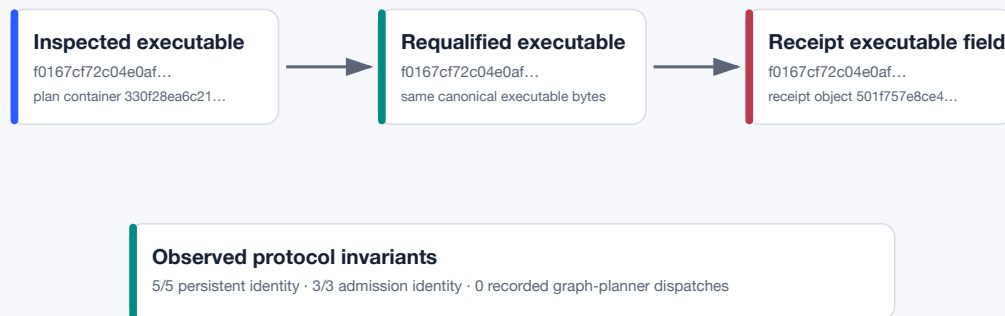
7 Historical C30 invariant-exercise method

C30 prospectively committed the lifecycle axes and a matched four-arm study in the author-controlled repository before the first pilot-evidence commit. The protocol commit is 2d54b8c; no external timestamp anchor is claimed. The present analysis uses its smaller, explicitly non-verdict-bearing mechanical invariant-exercise phase at the historical MENTU source epoch 40bd3f9. It does not measure the later Core extraction at f3d89e7.

The pilot used 1 bounded documentation objective in H-scaffold mode. A task-time planner constructed a strict DAG under an external authority envelope. Plan-only execution emitted canonical plan bytes. The operator inspected the plan and materialized a persistent recipe projection and parity record. A later invocation loaded the exact saved plan, rebuilt and requalified its executable against current state, admitted it, and invoked the runner without another graph-planner call. Read-only analysis then checked the plan, persistent recipes and prompt files, parity records, requalified admission bundles, receipts, call lanes, and sealed run manifests. For each persistent projection it reconstructs the normalized definition, contracts, effective policy, and canonical executable digest from source artifacts; the stored parity record is a cross-check, not the derivation. For each admitted execution it also rebuilds the requalified bundle identity from the admitted definition, actual prompt bodies, bound variables, contracts, and mechanically derived effective policy before comparing that digest with the receipt; a copied stored digest cannot satisfy this check. The planner-separation check has bounded, partly shared corroboration: the execution records state the dispatch count; every sealed call-lane record is bound to an admitted step activation and none is planner-classified; and a lexical audit of the pinned `runPlanned` implementation finds no planner emission. All three channels depend on recorded classification or one named code path; they are not independent instrumentation. The result is therefore zero *recorded* planner dispatches, not proof that every possible out-of-band planning path was observable.

Executable identity across the admission boundary

The same executable digest appears at inspection, requalification, and in the receipt binding.



Identity is about graph bytes—not deterministic node outputs or semantic correctness.

Figure 3: One representative historical C30 identity chain at 40bd3f9. The 3 admitted attempts carried distinct executable hashes. The receipt has its own identity while binding the requalified executable identity. This is pilot evidence, not a depiction of the later Core extraction.

The measurement pipeline has three layers. The frozen C30 analyzer remains the authoritative interpretation of the pilot corpus. Paper-local reproduction independently recomputes the identity and manifest invariants, reports source facts from committed blobs, and writes immutable JSON projections consumed by the manuscript and figures. It neither invokes the MENTU CLI nor writes to the measured stores or sealed run bundles. Finally, disposable seeded controls mutate copies only: one executable-bound contract field, one planner-lane record, and one manifest-listed byte. A separate control independently executes the committed current-Core admission-drift test in external build scratch. These controls assess detector sensitivity in two separately reported epochs; they are not C30 observations, a reliability coefficient, or complete I5 coverage.

The unit of an identity check is an addressable artifact transition, not an independent task trial. Qualified plan stages may be counted for projection parity; saved-plan execution attempts may be counted for inspection-to-receipt identity. They may not be treated as repeated outcome trials because the code and plan evolved during debugging. Human semantic inspection decisions are retained as human judgments and are not recoded as mechanically derived verdicts.

The pilot can falsify a protocol invariant: a receipt mismatch, an execution-time planner dispatch, or a manifest mismatch would be a concrete failure. It cannot estimate a static-versus-dynamic-versus-hybrid effect because the objective has no matched S or D arm. It also cannot establish planning amortization because there is no unchanged-state 3-repeat series.

8 Results and negative findings

The mechanically regenerated pilot summary is shown in Table 5. Its tabulated measurements and the principal counts in the surrounding prose are injected from generated result artifacts through the build context. Generated JSON remains authoritative where explanatory prose restates a small corpus descriptor.

The identity checks passed on the observed transitions: 5 of 5 persistent projections and 3 of 3 independently reconstructed requalified-bundle-to-receipt transitions retained executable identity. Saved-plan execution recorded 0 graph-planner dispatches. The sealed-bundle audit found all 54 expected leaves available, with zero missing and zero extra. It found 0 mismatches across those 54 file-digest checks and 0 mismatches across 3 aggregate checks derived from the same file sets. These observations support identity continuity and planner separation in this corpus. The transition funnel is explicit: five qualified projection attempts produced five identity checks; contemporaneous human inspection rejected v2 and v4 semantically, so three projections were selected for saved-plan execution. Those three execution attempts produced zero requalification refusals, zero admission refusals, three admissions, and three identity matches. The ratios are conditional transition checks, not end-to-end pipeline success rates. The planner result is additionally corroborated by 0 planner-classified and 0 unbound records among the sealed execution call lanes and by the pinned saved-plan source path. It does not imply that node-level model calls were absent, and none of these observations implies deterministic node output.

The historical paper-local sensitivity exercise detected 3 of 3 named faults. A mandatory contract-field mutation changed executable identity; an injected planner-lane record changed the planner

Check or observation	Type	Result	Context
Persistent-projection executable identity	Transition check	5/5	Qualified plan projections
Recomputed admitted-bundle-to-receipt identity	Transition check	3/3	Saved-plan executions
Graph-planner dispatches in saved-plan execution	Event count	0	3 saved-plan executions
Sealed-manifest file-digest mismatches	Leaf validation	0	54 file checks
Derived bundle-digest mismatches	Aggregate validation	0	3 bundle checks derived from those file sets
Distinct discovery repository heads	Corpus descriptor	3	5 plan stages
Mechanically qualified plans rejected semantically	Human judgment	2/5	Plan inspection
Qualified projections selected for saved-plan execution	Selection funnel	3/5	Versions v2 and v4 were not attempted after contemporaneous semantic rejection
Successful outputs requiring semantic correction	Human judgment	1/1	Completed outputs
Documented intervention decisions	Chronology	15	Full pilot history
Completed execution attempts	Chronology	1	3 chronological attempts

Table 5: Historical C30 invariant-exercise results. Every row derives from 1 H-scaffold objective ($n = 1$) carried through sequential development stages; the counts are not independent task trials.

count from zero to one; and one byte flip caused exactly 1 of 54 leaf digests and 1 of 3 derived bundle digests to fail. Separately, the current-Core sensitivity exercise detected 1 of 1 named faults: its admission-drift test refused a seeded captured-state mutation before execution. Input-tree fingerprints were identical before and after. These are positive controls for the named detectors, not additional pilot trials or proof of complete canonicalizer coverage. The historical payload-mutation control exercises one of the six mandatory F_χ classes (κ , contracts), with no historical equivalence, F_η -isolation, header-mismatch, or omission-witness fixture.

The 5 plan stages span 3 discovery repository heads, so the corpus is not wholly fixed-state: v1 maps to `c0adef6`, v2 to `37f9f2c`, and v3–v5 to `a828e91`. Thus the last three stages share a discovery head, while their plans and executable identities still differ. The 3 saved-plan executions are likewise chronological attempts, of which 1 completed the task. Reporting that fraction as a success rate would manufacture independence the design does not have. The 3 admission receipts also bind 3 distinct executable hashes: no executable graph was repeated, so the pilot contains no empirical comparison of output reproducibility for an unchanged graph.

The 54 leaf checks are nested inside 3 bundles inside one pilot corpus. They are not 54 independent trials, and no effective sample size of 54 is claimed. The byte-flip control illustrates the nesting: one changed leaf invalidated its leaf digest and the one derived bundle digest that contained it.

The strongest negative result concerns semantic validity. 2 mechanically qualified plans were rejected by human semantic inspection, and 1 mechanically successful output required post-run semantic correction. Mechanical qualification and step verification therefore did not establish plan adequacy or outcome correctness in the observed case. This is not an argument against qualification; it identifies the boundary of what those predicates certified.

The attempt chronology documents 12 task-planning dispatches. Complete telemetry exists only for the 5 saved qualified plans: 133,022 input plus 10,127 output tokens, totaling 143,149; 108,874 milliseconds elapsed; 490,151 request bytes; and 42,930 response bytes. The other documented dispatches lack complete telemetry, so these exact values are lower bounds for the full planning history rather than rounded estimates or a treatment effect. The 3 admitted attempts contain incomplete USD-labelled model-cost telemetry. The 2 attempts with nonzero telemetry have a US\$0.99 subtotal; the remaining 1 term is indeterminate because its recorded zero cannot be distinguished from missing telemetry. The provider pricing table used at collection time was not preserved, so no price recalculation or completeness claim is made. The same history contains 15 documented intervention decisions across provider, product, authority, plan, execution, and artifact-review stages. With 0 matched S observations and 0 matched D observations, none has a comparative interpretation. The number of unchanged-state repetition series is 0. The machine-readable comparative-verdict flag is `false`, read from the repository artifact `results/detector-sensitivity.json` (sha256 18584045...); the artifact is named because it sits outside any blind-review surface (finding BR9-007).

For the historical population, the evidence-adequacy vector is

$$\Gamma_E = (3, 3, 5, 3, 0, \emptyset, \emptyset, \omega_0) :$$

all three admitted runs have recorded host-boundary activity, three distinct admitted executable identities, five projections, three invocations in the reported I1 set, zero exercised header-refusal fixtures, zero exercised I4 drift categories, zero C30 I5 fixture classes, and the published observation profile ω_0 of Section 3. The first component is instrumentation evidence, not proof that every semantic effect was captured. Current-Core engineering tests and paper-local seeded controls are reported outside this vector.

Evidence boundary

Current-source, historical-pilot, and paper-control evidence remain separate.

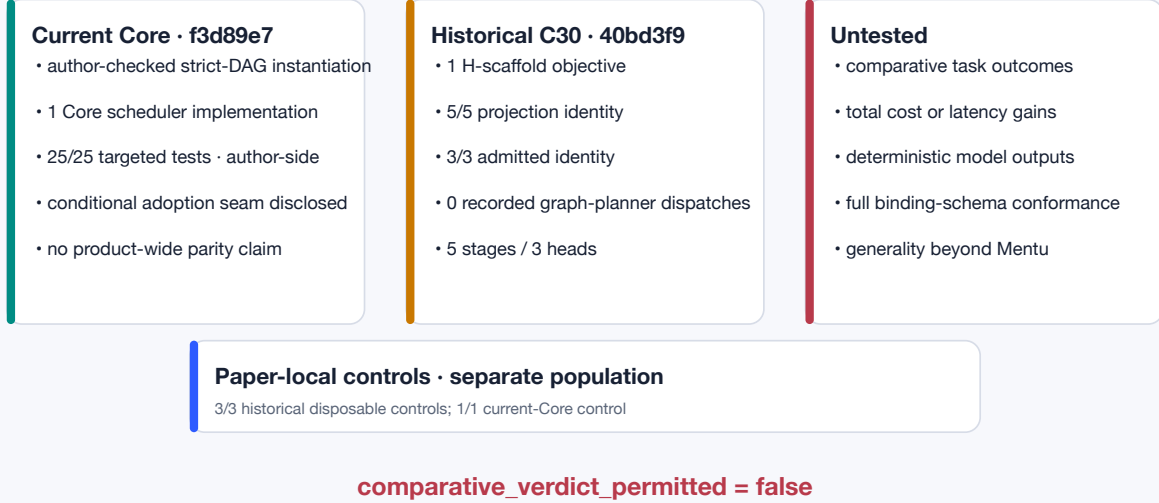


Figure 4: Claim boundary for v1.0. Observed protocol properties do not license unmeasured treatment effects.

9 Discussion: Object Graphs, Lifecycle Control, and Successor Epochs

The model admits a layered architectural reading. The executable task graph is the object-level course of action: it fixes effect-bearing nodes, contracts, and scheduling relations. The lifecycle state machine is a meta-level control structure: it records how that object was proposed, which predicates were established, when its identity became addressable, which concrete context authorized effects, and what evidence survived execution. “Meta-level” is used in an engineering sense; it does not ascribe self-awareness, privileged introspection, or semantic understanding to the runtime.

The separation changes the meaning of agent governance. MENTU does not make stochastic agent behavior deterministic. It makes an executable commitment inspectable and its authority to act attributable. Construction may remain exploratory before admission. Once an exact graph epoch has gained effect authority, however, a change to effect-bearing topology or scheduling semantics requires either a separately admitted successor epoch or an admitted outer transition system whose identity includes the planner and permitted transition relation. Dynamic construction and governed execution are therefore compatible without collapsing proposal into authority.

Execution evidence is not by itself a graph-update operator, and preservation is not learning. A cross-epoch extension could allow evidence and review to inform construction of a successor graph, but that successor would still require lowering, qualification, and admission. This represents recursive adaptation as an authorized lineage of graph epochs rather than silent mutation of an already

admitted graph. The present paper establishes the within-epoch substrate and the boundaries such an extension would have to preserve; it does not claim that cross-epoch learning is implemented or beneficial.

10 Threats, boundaries, and limitations

Construct validity. The model makes graph identity narrower than result reproducibility and both plan adequacy and outcome correctness. The versioned binding schema makes topology, node behavior, contracts, graph-declared scheduling, policy, and toolchain identity-bearing: effect-semantic fields enter F_χ , while the exact toolchain and schema header enters F_ν before payload canonicalization. Declared closure, payload equivalence/inequivalence, partition isolation, header matching, and omission probes are separate tests. This improves falsifiability but does not prove that the chosen schema captures every causally relevant fact. A model selector frozen by the executable is mandatory in executable identity; an independently resolved route, scheduler-build identity, provider runtime state, and authority state belong in the declared residual environment binding. The pilot binds declared environment constituents through qualification and admission records but does not compute one aggregate environment digest, so it cannot report $h_{\eta,1} = h_{\eta,2}$. A collision-resistant digest is evidence about identical encoded objects, not about the adequacy of the encoding.

Internal validity. The C30 corpus records a development pilot whose implementation, authority configuration, and plans changed between stages. Its 3 repository heads make that evolution visible, but they also prevent a fixed-state outcome interpretation. Human semantic rejections and the post-run correction are important negative evidence, yet they were made by the author-operator during development rather than by blinded raters. Some telemetry, including human inspection time and a complete execution-token account, was not available. The pilot challenged zero drift classes; drift refusal remains prospectively specified but empirically untested in C30. The seeded current-Core drift control is separately labeled engineering-test evidence and does not change that denominator.

External validity. The empirical material is confined to the observed H-scaffold objective. The current implementation mapping is confined to MENTU as the author-checked architectural instantiation. Its admitted profile accepts strict DAGs, whereas the abstract model allows a wider graph class. The mapping therefore demonstrates one coherent realization, not cross-platform generality or comparative rank. Systems with runtime topology mutation, open-ended tool arbitration, or different authority models may need multiple admitted epochs or a different abstraction.

Comparative boundary. There are no matched S or D observations, no H-direct control, and no unchanged-state repetition series. The paper cannot infer lower total cost, lower latency, fewer regressions, reduced intervention, improved task success, or planning amortization. Static authoring effort and human inspection effort would need explicit measurement in any such comparison.

Implementation boundary. The current admitted profile has a closed Core definition, qualification and admission primitives, activation evidence, and one Core scheduler. General recipes and interpreted workflows remain compatibility profiles with different authority or runner contracts.

The source audit does not enumerate every non-graph effectful product operation, so it makes no product-wide coverage percentage. It also finds that persistent generated projections can re-enter legacy recipe dispatch without explicit adoption. That cross-profile transition is a real adoption boundary; it neither erases the admitted Core nor licenses a claim of uniform admission.

Scholarly and review boundary. A bounded set of cited primary records from earlier cycles has had its source identity independently reconciled and its claim-specific use explicitly dispositioned. The v0.9.5 exact-input return passed its mechanical isolation gate, but independent audit found five copy-contaminated RFC author fields, a 34-of-37 honest citation-entity crosswalk, and one mutable-versus-pinned source drift. Its literature coding is bounded discovery evidence and its blind review is major-revision input, not certification. This revision therefore also records that v0.9.6 failed its full raw gate and v0.9.7 failed independent trace audit despite a green structural validator. The latter return’s 43/43 cite-key identity crosswalk does not certify full rendered bibliography metadata, and its RFC 2748 record fails the protocol’s first-author condition. This revision therefore requires a fresh exact-input, trace-audited run; no priority, exhaustive-coverage, or novelty claim is licensed.

Privacy and reproducibility boundary. The public package uses digests, summaries, committed blob identities with path-and-line locators, and path-neutral scripts. It excludes credentials, transcripts, raw session exports, private filesystem paths, and sealed run-bundle contents. This protects private material but means that some sealed evidence must be verified by an authorized local reproducer rather than redistributed. Bare hashes of short source lines are deliberately excluded because they are guessable and provide no confidentiality. Timestamp-suffixed run identifiers and the pilot’s single-operator telemetry may support work-pattern inference. They are explicitly not privacy controls. The current artifact remains internal. A public release must either publish a pseudonymized projection whose invariants still pass or record the author-operator’s explicit consent to the retained activity data. Before a future study records other humans’ authoring or inspection time, its protocol must document an ethics determination and a consent procedure; the present pilot is single-operator and supplies neither for a multi-participant study.

11 Prospectively committed matched-study protocol

The adjudicating study was specified in the author-controlled C30 record before the pilot evidence and is restated here to separate future inference from pilot interpretation. No external registration timestamp is claimed. It freezes at least 30 objectives spanning at least 3 task families and 3 size strata. Each objective is executed in S, D, H-scaffold, and H-direct arms on resettable repository fixtures. The floor of 30 objectives is an instrument-operability threshold, not a power justification. Before inferential execution, a precision analysis will set the final corpus size from the target width of the paired verified-completion interval. Backend, execution model, authority envelope, token and cost ceilings, and mechanical verification contracts are held constant within objective.

Arm order is randomized within objective, and every attempt is retained, including pre-dispatch refusals and execution failures. Static recipes are authored before randomized execution begins; their human or model authoring time is measured rather than treated as zero. H-scaffold and H-direct share the staged freeze policy so their contrast isolates authoring method more cleanly than a comparison with D. S and D provide the persistent and continuous-invocation reference policies.

The primary comparative endpoint is verified task completion. Exact executable identity, planner-dispatch count by lane, pre-effect drift refusal, and manifest integrity are protocol-conformance outcomes. Secondary comparative endpoints are intervention count, elapsed time, model cost, planning tokens, authoring time, inspection time, and typed refusal rate. A hierarchical analysis tests the primary endpoint first; secondary intervals use a Holm family-wise adjustment and unadjusted estimates are labeled exploratory. Every eligible first attempt—including a refusal or failed execution—enters the primary analysis and is followed by 3 unchanged-state invocations when the arm remains operationally eligible. Reuse and amortization are reported both unconditionally and, separately labeled, conditional on an admitted first execution. Outputs may vary; the executable identity requirement does not.

Analysis is paired by objective. It reports arm-specific distributions, paired differences, medians, and bootstrap confidence intervals rather than promoting pilot counts to population estimates. Refusals are safety outcomes, not silently recoded task successes or failures. Deviations, exclusions, and missing telemetry remain visible. The detailed execution and adjudication rules appear in the appendix.

12 Conclusion

Static, dynamic, and staged-hybrid agent execution need not imply separate runtimes. They can be represented as policies over when a graph is constructed, where it freezes, how it is authored, and which substrate executes the admitted result. That abstraction is useful only if it preserves the boundaries among executable identity, environment identity, output reproducibility, plan adequacy, outcome correctness, and authority.

MENTU operationalizes the model’s central separation through an extracted Core with a public API that owns the closed executable, qualification and admission primitives, activation evidence, and strict-DAG scheduler. Engine adapters supply host effects, while named compatibility profiles retain different contracts. The audit also exposes the next coherent hardening step: persistent generated projections need an explicit adoption transition before they can enter a different assurance contract.

The historical C30 checks passed on their observed identity and planner-lane transitions, while preserving semantic failures that mechanical qualification did not catch. Seeded controls show sensitivity to 3 of 3 named historical faults; a separate current-Core control detected 1 of 1 current fault. Neither result supplies a comparative verdict, aggregate environment-identity result, unchanged-graph output repeat, or complete I5 conformance claim.

The next scientific steps are therefore bounded: literature-check and blind-review exactly this frozen internal PDF and supplement in isolated contexts, then run the prospectively committed matched study. Until those gates close, the contribution is a testable lifecycle model, an author-checked architectural instantiation, an implementation-boundary finding, and a bounded invariant exercise—not a claim of novelty, universality, comparative rank, status, or superiority.

Appendix: Evidence, claims, and matched-study detail

A. Provenance and evidence grades

Table 6 names the manuscript’s load-bearing source classes. Repository-local paths are resolved by the reproduction wrapper; no private absolute path is part of the paper source or public package. Input digests and the complete file-level mapping are recorded in `SOURCE-MAP.md` and the generated evidence manifest.

Source	Pinned identity	Permitted use
C30 prospective protocol	2d54b8c	Frozen definitions, predictions, falsification rules, and matched-study design
C30 pilot corpus	867c95f	Read-only historical invariant-exercise observations and negative findings
MENTU historical source	40bd3f9	C30-era source corroboration only
MENTU current source	f3d89e7	Current Core, adapter, scheduler, admission, compatibility-profile, and projection-adoption facts
Structural Waste	fe02777	Methodological discipline only; no substantive theory transfer
Claude Science exports	External raw inputs, excluded from Git	Provenance of failed attempts and bounded Operon returns; raw session frames are not release evidence

Table 6: Source roles. Committed source and frozen reproducible analysis are internally high-assurance author-produced evidence; sealed hash-checkable local artifacts are access-restricted; dirty worktree and vendor-reported observations are non-load-bearing. These grades do not claim independent public replication.

The external research exports are retained outside the repository. The first export referenced `ground_truth.json`, which was unavailable; no reconstruction is treated as equivalent. The earlier 29 July return exported an older manuscript-construction task and never received the target PDF or public supplement. The first 30 July return matched the task but failed the isolation and exact-input contract; its canonical artifacts remain preserved as advisory evidence. The subsequent v0.9.3 rerun passed the reported memory-isolation and exact-two-file input gate and is preserved byte-for-byte under `operon-v093/`. Its green attestation records the declared process conditions, not substantive correctness; the blind review recommended major revision. The later exact-input v0.9.4 return is preserved byte-for-byte under `operon-v094/`; independent audit overrides its coordinator’s green narrative because the frozen raw-return validator fails. Downloaded source maps were replaced by a repository-owned corrected map rather than overwritten. The v0.9.5 return is preserved under `operon-v095/`. Its isolation gate passes, while independent audit rejects literature certification because five RFC author records are copy-contaminated and the citation registry cannot be honestly crosswalked one-to-one to all cited entities. Its 44-finding blind review remains major-revision evidence. The v0.9.6 return is preserved as a failed raw-gate epoch with 21 advisory findings, six major. The v0.9.7 raw files are also preserved byte-for-byte, but their green structural result is overridden for release purposes by the private session trace: Contexts A and B overlapped, both read undeclared text derivatives, automated reviewers crossed the claimed boundary, and the same Context A was resumed to repair a failed full-validator result. Its 21 findings remain advisory repair inputs; the run supplies no release authorization. Uncommitted terminal and GUI evidence

is recorded only as internal provenance.

The v0.9.9 certification program ran six sealed protocol epochs (r1–r6), and every one ended honestly without producing a certifiable review. r1 and r2 stopped in preflight on frozen vocabularies that lacked a term for something the platform emitted. r3 reached adjudication and returned non-compliant on six codes. r4 stopped on a contradiction between two pinned protocol documents; r5 on a supplied export of a session that had never finished. r6 with custody preflights closing every previously discovered gap reached adjudication and returned noncompliant on seven producer-conduct codes, principally the divergence of producer self-describing logs from the platform-represented trace: fifty-seven logged network requests against zero represented network events, and one hundred four file-open claims contradicted by the represented event scan. Each epoch’s raw tree, toolchain, and stop or void record is preserved byte-for-byte under its own lineage (**provenance/OPERON-V099*.md**, **operon-v099*/**, **operon-certification-v099*/**). The r6 blind reviewer satisfied its own distinguishing constraints, and its seven findings are dispositioned in the supplement as uncertified advisory review; findings BR9-001 through BR9-007 are addressed or scheduled in this candidate. By dated author decision (**provenance/RELEASE-BOUNDARY-DECISION-2026-08-14.md**), a certified review is no longer a release precondition; disclosure of this six-epoch record is. The repeated adjudication result that unverified producer self-reports diverge from platform-represented traces is itself evidence for the position this paper takes on trace-derived, rather than self-attested, process records.

B. Manuscript-facing claims ledger

This manuscript and its co-reviewed reproducibility supplement are authoritative for public-facing claims. The machine-readable **spec-map.json** and full **CLAIMS-LEDGER.md** are internal audit meta-data and must not broaden this reviewed surface. Table 7 is the complete manuscript-facing projection of the material claim classes.

C. Implementation and assurance-contract map

Table 8 separates shared execution machinery from policy-specific assurance and topology contracts.

ID	Class	Claim	Status
AGR-AD-01	architectural definition	Construction time, freeze boundary, authoring method, and execution substrate are separately specified coordinates over a constrained feasible region.	Defined here; precedents positioned without novelty claim
AGR-FP-05	formal implication	Multiple policy configurations can target one executable IR and substrate without sharing an authoring path.	Conditional on the model
AGR-IF-11	implementation	MentuExecutionGraphCore owns the admitted definition, lifecycle, qualification, admission, activation evidence, and DAG scheduler; engine adapters supply effects.	Current-head source plus 25/25 targeted author-side tests; no product-wide cross-origin execution claim
AGR-IF-12	fact		
AGR-IF-13	implementation fact	Persistent generated projection can re-enter ordinary recipe discovery without an explicit adoption transition.	Current cross-profile transition boundary
AGR-IF-14	implementation fact	MENTU is the author-checked architectural instantiation of the strict-DAG admitted profile.	Profile-scoped synthesis; not a comparative rank
AGR-PO-01	pilot observation	Saved-plan and persistent-projection identity held on all observed artifact transitions.	Noncomparative pilot evidence
AGR-PO-02	pilot observation	Saved-plan execution contained no recorded graph-planner dispatch in the observed attempts.	Noncomparative pilot evidence
AGR-PO-03			
AGR-PO-06	pilot observation	Mechanical qualification did not establish plan adequacy or outcome correctness in the observed case.	Negative result retained
AGR-PO-07			
AGR-EX-01	excluded claim	Staging improves success, total cost, latency, intervention, maintenance, or regression outcomes.	Not permitted
AGR-AD-06	architectural definition	Unadmitted runtime topology or scheduling mutation is not another policy over one frozen graph epoch.	Scope boundary retained

Table 7: Claim classes and current dispositions. “Supported” is scoped to the named evidence; it is not a cross-platform generality claim.

Layer	Admitted graph profile	Recipe compatibility profile	Interpreted workflow profile
Executable form	Core ExecutionGraphDefinition through checked engine adapter	General SequenceDefinition	Validated JavaScript program
Authority entry	Core qualification and admission before sole admitted runner	Raw compatibility initializer with legacy authority	Separate WorkflowRunner contract
Scheduling	One Core ExecutionGraphScheduler for the admitted DAG	DAG-shaped recipes may use the Core scheduler through the engine adapter; raw compatibility routes use their selected SequenceRunner contract, without a paper-wide scheduling-convergence claim	Interpreter-defined transitions
Evidence	Qualification, receipt, activation, call-lane, and run evidence	Sequence/run evidence under the selected recipe contract	Workflow journal and budgets; no admitted-meta-runtime claim here
Cross-profile transition	Projection stays in-profile or is explicitly adopted	Generated projection can currently be rediscovered here without explicit adoption	Not classified as a frozen inner-graph policy

Table 8: MENTU runtime surfaces and assurance contracts at commit **f3d89e7**. The admitted Core is the named instantiation; compatibility profiles are named rather than treated as missing pieces of that Core.

D. C30 pilot interpretation rules

The paper-local reproduction applies the following rules without rewriting the frozen C30 result:

1. Independently load each persistent recipe and prompt set, reconstruct its normalized definition, contracts, effective policy, and canonical executable digest, then cross-check the stored parity record.
2. Compare the inspected executable identity with the requalified admission bundle and receipt for every admitted saved-plan execution.
3. Count graph-planner dispatches only in the saved-plan execution invocation; admitted node-model calls belong to a separate execution lane.
4. Recompute every available sealed-manifest entry and count mismatches without copying private contents.
5. Count distinct discovery repository heads across plan stages to distinguish evolving debugging stages from unchanged-state repeats.
6. Preserve human semantic rejection and correction records as human judgments; do not transform them into mechanical analyzer outputs.
7. Set the comparative verdict to false whenever matched S and D arms or the registered repetition series are absent.

An identity pass supports only the relevant artifact transition. A pre-dispatch refusal is a safety

outcome. A completed step is not semantic certification. Sequential attempts are not independent trials. These rules are enforced in the claims ledger as well as in prose.

E. Prospectively committed matched-study execution

Experimental unit and corpus. The experimental unit is one frozen objective evaluated under all four policy arms. The corpus contains at least 30 objectives across at least 3 task families and 3 size strata. Objectives, eligibility rules, task-family labels, size labels, fixture-reset procedures, and verification contracts are frozen before the first randomized arm is executed.

Arms.

S: a persistent recipe and prompts exist before the experimental task invocation; graph-planner dispatch during execution is zero.

D: one task-time planner dispatch constructs, qualifies, and executes the graph continuously without an operator-visible frozen plan.

H-scaffold: a scaffold-method task-time planner creates an addressable plan; a later invocation requalifies and executes those exact bytes.

H-direct: the same staged boundary as H-scaffold with the direct authoring method, isolating staging from authoring style.

Controls and randomization. Each objective begins from a resettable repository fixture. Execution backend and model, authority limits, token and cost ceilings, sandbox and permission posture, expected-change scope, and mechanical verification remain matched. Arm order is randomized within objective. Static artifacts are authored and frozen before randomized execution; authoring labor and model usage are recorded. Every invocation receives a stable objective identifier, arm, order position, fixture identity, repository identity, authority-envelope identity, and artifact lineage.

Attempt accounting. All attempts remain in the record. Eligibility requires valid authority, available credentials and provider, no disqualifying fixture drift, and arrival at qualification. Infrastructure refusals remain visible but are excluded from task-success denominators according to the frozen rule. Human edits, retry decisions, authority changes, and plan regenerations are interventions. Mechanical drift refusals are typed separately. The analysis reports both the intention-to-run history and the eligible-attempt subset.

Measurements. Protocol measures include planner dispatch count and lane, request and response bytes, input and output tokens, elapsed planning time, candidate source identity, executable identity, qualification identity, receipt identity, manifest integrity, and pre-effect drift behavior. Operational measures include verified completion, changed paths, execution duration, model cost, interventions, typed refusals, and reusable artifacts. Human authoring and inspection duration are timed explicitly. CLI-return latency is separated from runner duration so post-run extraction cannot be mistaken for execution latency.

Repetition and outcomes. After a successful first execution, each arm receives 3 unchanged-state repetitions from an equivalent reset fixture. Graph identity is expected to remain exact where the policy freezes an addressable artifact; model outputs are allowed to vary. The primary falsification outcomes are any staged receipt identity mismatch, any graph-planner dispatch during saved-plan execution, or any detected drift that reaches effectful execution. Comparative outcomes remain descriptive until the complete matched design is analyzed.

Analysis and reporting. Results are paired by objective. The report provides arm-level distributions, paired differences, medians, bootstrap confidence intervals, missingness, and typed refusal counts. H-scaffold versus H-direct estimates an authoring-profile contrast under a shared staging policy; comparisons involving D or S change additional lifecycle dimensions and are interpreted accordingly. No outcome is promoted from the pilot. Deviations from the frozen protocol receive dated amendments, and null or adverse findings remain in the release artifact.

F. Operon release gate

The internal manuscript becomes eligible for a public preprint only after:

1. input-bounded discovery, followed by full-text or identifier checks, across agent frameworks, planner-executor systems, graph runtimes, planning, content-addressed builds, durable execution, provenance, and admission, in a separately closed review-domain session;
2. a verified prior-art registry and literature matrix;
3. blind review of exactly the internal PDF and public supplement in a second separately closed review-domain session, with a digest-bound deterministic text projection of that same PDF permitted only as a navigation aid, and with no network, Context A artifact, internal claims ledger, or spec map as input;
4. detached derivation over both closed-session exports, scanning each root frame and its complete descendant closure for undeclared inputs, cross-domain references, and prohibited effects;
5. explicit disposition of novelty, terminology, formal-soundness, empirical-overclaim, citation, and reproducibility findings; and
6. an integrity build in which cited and bibliography sets match and no unverified identifier remains;
7. author-owned publication clearance binding a public licence, a stable release locator, and explicit consent for retained timestamp-bearing activity data.

Dated status. The eligibility rule above records the gate as designed. By the author’s boundary decision of 2026-08-14, certification ceased to be a release precondition (disclosure of the epoch record, incorporation of the uncertified blind-review dispositions, and the explicit release act are); the release act occurred on 2026-08-22 under Decision A1, with the machine gate’s refusal of this candidate preserved undisturbed in the repository record rather than amended away.

The machine gate treats the return as a closed evidence layout. It binds canonical prompt and preflight bytes, proves the declared text projection against the authoritative PDF, rejects undeclared local reads and all blind-domain network access, recomputes manuscript citation locations, and requires auditable receipts for source resolution and finding disposition. A review domain is the complete frame graph reachable from one exported root, not only its initiating frame. Declared reviewer or critic descendants inside that closure are therefore visible parts of the review computation; an edge, read, identifier, or artifact crossing from the literature domain into the blind domain is disqualifying. After both producers terminate, a separate Codex-side extractor reads

and hashes the actual exports and derives the lifecycle, tool, artifact, and information-flow record. The producer cannot submit a substitute audit JSON or run the certifying validator. The raw exports remain private and outside the public package; the derived audit records only path-neutral identities, counts, locators, digests, and a scoped result.

The result is explicitly relative to events represented in the supplied exports. Because those exports are not platform-signed completeness proofs, unobservable host controls remain unknown: an absence in the export is not promoted into proof that omitted platform state could not exist. Review independence, observable non-interference, artifact consistency, scholarly correctness, and author release authority are separate decisions. Release validation, the reviewed internal rebuild, and public packaging run from one frozen source snapshot. These controls establish byte identity, provenance, and review traceability; they do not mechanically prove scholarly correctness or semantic support. Search breadth and literature-matrix coding likewise remain attested scholarly process: the gate binds the returned artifacts but cannot infer exhaustive coverage or semantic correctness from their presence.

Operon findings are review inputs, not automatic authority. The v0.9.5 exact-input return supplied a substantive major-revision review and passed its frozen process gate, but its copy-contaminated and non-bijective literature registry cannot authorize release. The v0.9.6 return failed its raw gate. The v0.9.7 return demonstrates why self-attested isolation is also insufficient: its raw validator is green while trace audit is red. The v0.9.8 preflight adds the converse lesson: an unverifiable audit must stop before review rather than be completed by assertion. This revision changes the PDF, supplement, review protocol, and build projection. Each citation and revision remains subject to repository-owned verification and disposition. The r6 epoch then reached adjudication and returned noncompliant on seven producer-conduct codes; the six-epoch record is disclosed in Appendix A. By the dated author decision of 2026-08-14, release no longer waits on a certified review; it requires the disclosed record, the incorporated r6 uncertified-review dispositions, and the author's explicit release act.

References